

SUMMARY

Computational biologist and AI scientist with expertise in machine learning, multimodal omics, spatial biology, and translational oncology. Experienced in developing AI-driven methods for biomarker discovery, patient stratification, and biomedical knowledge extraction through collaborations with clinicians, pharmaceutical partners, and international research institutes. Strong background in deep learning, LLMs, GPU computing, and scalable bioinformatics pipelines.

CORE SKILLS

Machine Learning & AI: Generative AI (LLMs, Multi-agent systems, RAG, LangChain, LangGraph, LangSmith), Deep Learning (PyTorch), Transformer Architectures, Statistical Modeling, Predictive Analytics, Clinical ML.

Bioinformatics & Computational Biology: Single-cell & Spatial Omics, Multi-omics Integration, Drug Target Identification, Oncology & Hematology Research.

Software & Infrastructure: Python, R, Nextflow, Snakemake, Docker, Cloud Platforms (AWS, GCP), High-Performance Computing (HPC), System Benchmarking.

Project Leadership: Stakeholder Management, Technical Mentorship, Rapid Prototyping, Cross-functional Collaboration.

METRICS

Research Impact: 13 publications, 2 provisional patents, 100+ students mentored, International collaborations with Broad Institute, Dana-Farber, Standard BioTools, and University of Galway

WORK EXPERIENCE

University of Galway

Galway, Ireland

Research scientist - Postdoc

06, 2026 – Now

- Developed and deployed end-to-end clinical machine learning models for oncology.
- Architected agentic AI systems to automate research workflows by integrating disparate databases into analytical pipelines.
- Delivered timely clinical biostatistical analyses and technical insights for external partners.
- Developed and instructed genomic data analysis coursework for BSc and MSc students.

Broad Institute of MIT and Harvard/Dana Farber Cancer Institute

Boston, MA

Visiting Researcher, Medical Oncology

02, 2024 – 11-2025

- Led the computational analysis for multi-modal oncology research, partnering with clinicians to analyse spatial biology data for clinical applications (Paper Submitted).
- Managed a 400-sample scRNA-seq study using UK Biobank data; built GCP-based genomic pipelines with Python and Docker to process high-dimensional datasets.
- Developed statistical and ML models for CAR-T patient stratification, supporting myeloma drug discovery and biomarker evaluation (provisional patent).
- Partnered with StandardBioTools to integrate 3D imaging and GPU acceleration, reducing pipeline image registration times by 90%.
- Deployed a U-Net model for image denoising, optimizing GPU-accelerated training and inference workflows.

University of Galway

Galway, Ireland

Researcher

09, 2021 – 11, 2025

- Analyzed a multi-center 522,000+ cell dataset to model immune cell exhaustion, identifying novel therapeutic targets currently in in vivo validation.
- Developed a computational framework using maximum-flow network theory and multi-objective genetic algorithms to prioritize key biological drivers.
- Architected a containerized Nextflow pipeline automating high-throughput single-cell processing, transcriptional profiling, and cellular interaction inference.
- Co-developed an AI system using LLMs, SQL, and RAG to automate unstructured data extraction from literature with tool and code execution.
- Supervised and mentored 100+ students; taught genomic data analysis coursework, and cloud computing (AWS).

CloudCix

Cork, Ireland

Tester

01, 2023 – 06, 2023

- Benchmarked deep learning training across multi-GPU cloud instances using PyTorch and CUDA to evaluate and optimize hardware acceleration.

Ancona University Hospital

Ancona, Italy

Research Assistant, Immunology

07, 2019 – 09, 2021

- Analyzed clinical and genetic datasets for over 100 immunology patients to evaluate disease risk factors.
- Conducted safety and efficacy analyses for CSL Behring intravenous pharmaceutical products.
- Collaborated with clinical teams to translate complex biological datasets into research insights.

Queen's University Belfast

Belfast, UK

Intern

02, 2020 – 08, 2020

- Executed virtual screenings of 500+ small molecules, utilizing molecular dynamics and protein-protein interaction networks.

EDUCATION

University of Galway

PhD, Genomics Data Science

Marche Polytechnic University

Master's Degree, Molecular and Applied Biology

Grade: 110L/100 (4.0 GPA)

Marche Polytechnic University

Bachelor's Degree, Biological Sciences

Grade: 108/100 (3.9 GPA)

Galway, Ireland

09, 2021 – 11, 2025

Ancona, Italy

2018 – 2021

Ancona, Italy

2015 – 2018

CERTIFICATIONS

Anthropic: Model Context Protocol (MCP), Agents Skills (**ongoing**)

Google: GenAI Intensive Course (**04/2025**)

NVIDIA: Building Transformer-based NLP Applications (**11/2024**)

NVIDIA: Fundamentals of Deep Learning (**10/2024**)

CITI Program: Good Clinical Practice (GCP) for Clinical Research (**02/2024**)

CITI Program: Human Research (**02/2024**)

University of Galway: Project Management in the Real World (**11/2024**)

AWARDS

Thomas Crawford Hayes Travel Fund winner, **€2300**.

Longhack & VitaDAO 2023 hackathon Winning team, Team Leader, **€1000 + 1000 VITA-Coin**

1st prize for the Decisive Data Podium Session at the CSE Research Day, **€200**

PROJECTS

miRKatAI: Agentic AI for miRNA research, Google Gen AI, SQL, Prompt engineering, GroundSearch, Streamlit

Breast Cancer Chatbot: Assistant to analyze clinical/molecular data, Bioinformatics Workflows, AI Agents,

LLM, Autoencoders

AI Maria: Fine-tuned LLM for Myeloma Research Assistant, Transformers, Fine tuning, RoBERTa

LANGUAGES

Language: English (C1 - Full Professional Proficiency), Italian (Native), Spanish (B2)

PUBLICATIONS

- [1] Karen Guerrero-Vazquez et al. "miRKatAI: An Integrated Database and Multi-agent AI system for microRNA Research". In: (2025). DOI: 10.48550/ARXIV.2508.08331. URL: <https://arxiv.org/abs/2508.08331>.
- [2] Verga, **Jacopo U.** and Foster, Kane and Cordas Dos Santos, David M. et al. "Bone Marrow Spatial Signatures Predict Survival Outcomes and Toxicities after CAR-T Therapy in Patients with Relapsed/Refractory Multiple Myeloma". In: *Cancer Discovery - Under Review* (2026).
- [3] Yoshinobu Konishi et al. "Bone Marrow Spatial Signatures Predict Survival Outcomes and Toxicities after CAR-T Therapy in Patients with Relapsed/Refractory Multiple Myeloma". In: *Blood* 144.Supplement 1 (Nov. 2024), pp. 590–590. ISSN: 1528-0020. DOI: 10.1182/blood-2024-204144. URL: <http://dx.doi.org/10.1182/blood-2024-204144>.
- [4] Verga, **Jacopo U** et al. "NK Cell States in Multiple Myeloma: Pathways to Novel Immunotherapies". In: *Blood* 144.Supplement 1 (Nov. 2024), pp. 6857–6857. ISSN: 1528-0020. DOI: 10.1182/blood-2024-210297. URL: <http://dx.doi.org/10.1182/blood-2024-210297>.
- [5] David Cordas dos Santos et al. "P-203 Ability To Perform Spatial Transcriptomics in FFPE Decalcified Bone Marrow Samples of Patients With Precursor Myeloma". In: *Clinical Lymphoma Myeloma and Leukemia* 24 (Sept. 2024), S154–S155. ISSN: 2152-2650. DOI: 10.1016/s2152-2650(24)02106-2. URL: [http://dx.doi.org/10.1016/S2152-2650\(24\)02106-2](http://dx.doi.org/10.1016/S2152-2650(24)02106-2).
- [6] Verga, **Jacopo U** et al. "Genomic technology advances and the promise for precision medicine". In: *Therapeutic Drug Monitoring*. Elsevier, 2024, pp. 355–371. ISBN: 9780443186493. DOI: 10.1016/b978-0-443-18649-3.00007-0. URL: <http://dx.doi.org/10.1016/B978-0-443-18649-3.00007-0>.
- [7] Verga, **Jacopo U** et al. "A Systems Biology Approach Reveals the Endocrine Disrupting Potential of Aflatoxin B1". In: *Exposure and Health* 16.2 (May 2023), pp. 321–340. ISSN: 2451-9685. DOI: 10.1007/s12403-023-00557-w. URL: <http://dx.doi.org/10.1007/s12403-023-00557-w>.
- [8] Zachary Boswell et al. "In-Silico Approaches for the Screening and Discovery of Broad-Spectrum Marine Natural Product Antiviral Agents Against Coronaviruses". In: *Infection and Drug Resistance* Volume 16 (Apr. 2023), pp. 2321–2338. ISSN: 1178-6973. DOI: 10.2147/idr.s395203. URL: <http://dx.doi.org/10.2147/IDR.S395203>.
- [9] Maria Giovanna Danieli et al. "Common Variable Immunodeficiency in Elderly Patients: A Long-Term Clinical Experience". In: *Biomedicines* 10.3 (Mar. 2022), p. 635. ISSN: 2227-9059. DOI: 10.3390/biomedicines10030635. URL: <http://dx.doi.org/10.3390/biomedicines10030635>.
- [10] Verga, **Jacopo U** et al. "Integrated Genomic and Bioinformatics Approaches to Identify Molecular Links between Endocrine Disruptors and Adverse Outcomes". In: *International Journal of Environmental Research and Public Health* 19.1 (Jan. 2022), p. 574. ISSN: 1660-4601. DOI: 10.3390/ijerph19010574. URL: <http://dx.doi.org/10.3390/ijerph19010574>.
- [11] George S. Hanna et al. "Contemporary Approaches to the Discovery and Development of Broad-Spectrum Natural Product Prototypes for the Control of Coronaviruses". In: *Journal of Natural Products* 84.11 (Oct. 2021), pp. 3001–3007. ISSN: 1520-6025. DOI: 10.1021/acs.jnatprod.1c00625. URL: <http://dx.doi.org/10.1021/acs.jnatprod.1c00625>.
- [12] Veronica Pedini et al. "Incidence of malignancy in patients with common variable immunodeficiency according to therapeutic delay: an Italian retrospective, monocentric cohort study". In: *Allergy, Asthma & Clinical Immunology* 16.1 (June 2020). ISSN: 1710-1492. DOI: 10.1186/s13223-020-00451-z. URL: <http://dx.doi.org/10.1186/s13223-020-00451-z>.
- [13] Maria Giovanna Danieli et al. "A Case of COVID-Associated Inflammatory Bowel Disease with CTLA-4 Mutation Treated with Abatacept". In: *Archives of Clinical and Medical Case Reports* 03.06 (2019). ISSN: 2575-9655. DOI: 10.26502/acmcr.96550124. URL: <http://dx.doi.org/10.26502/acmcr.96550124>.